
1. Authentication & Security

- The documentation mentions **two-layer authentication** (SSL certificate exchange + transport key-based auth), but our environment supports **Basic Auth**.
 - Will MTN allow Basic Auth for initial integration, or must Fiorano ESB handle certificate-based mutual TLS (mTLS)?
 - If mTLS is mandatory, can MTN provide guidelines for importing and managing certs in a Java-based ESB?
- What is the **certificate renewal policy** (expiry period, notification window, renewal process)?
- Is the **private key exchange** process secure and compliant with banking standards?
- Can we test against **self-signed certificates** in UAT or do we need CA-signed even in non-production?
- Will we receive **separate credentials/certs for UAT and production**?

2. Environment & Connectivity

- IP whitelisting is required:
 - Can MTN confirm the list of IPs for UAT and production endpoints?
 - Is there a separate **sandbox API base URL** for testing?
- Are there **firewall-level failover IPs** or alternate hosts if the primary endpoint is down?
- What is the **expected latency** per request for real-time settlement?

3. API Protocol & Compatibility

- Are all endpoints **strictly XML** or can we send requests in JSON? (The document specifies XML, but Fiorano supports multiple formats.)
- Do the APIs require **specific XML namespaces** exactly as in the XSDs?
- Are **schemas (XSDs)** available for all API versions mentioned (v1.0, v2.0, etc.)?
- How strict is the **element ordering** in XML requests?

4. Transaction Management

- In **withdrawal flows**, the `paymentinstruction` initially returns **PENDING**:

- What is the maximum wait time before the `paymentinstructionresponse` is sent?
 - If no response is received after a certain time, should we retry or query status using `gettransactionstatus`?
 - For **duplicate detection**, what criteria does MTN use to identify a duplicate transaction (transaction ID, bank ID, timestamp, amount)?
 - For **SUSPENDED** transactions in deposits, do we have an API to retry or must this be done manually by MTN support?
 - How are **partial failures** handled (e.g., deposit credited in MTN but failed in bank ledger)?
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5. Error Handling & Logging

- Can MTN provide a **full list of error codes and messages** beyond what's in the doc, including operational ones (timeouts, malformed XML, auth failures)?
 - For validation failures, does the API return a **machine-readable code** or just a message string?
 - Are error messages **localized** or always in English?
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6. Performance & Throughput

- Is there a **rate limit** per second/minute for any of the APIs?
 - What is the **maximum payload size** for deposit and withdrawal requests?
 - Are there **batch APIs** for deposits, or must each deposit be sent individually?
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7. Compliance & Audit

- Since escrow transactions are regulated, is MTN **digitally signing** the responses for audit integrity?
 - Are there **unique reference rules** (e.g., `banktransactionid` must be globally unique for all time)?
 - What is the **minimum retention period** for transaction data in MTN's system?
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8. Testing & Go-Live Process

- Will MTN provide **mock MSISDNs, agent IDs, and test scenarios** for UAT?

- Is there a **switch-over checklist** for moving from UAT to production?
 - How will MTN notify us about **production changes** to APIs (e.g., version deprecation)?
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Potential Concerns from Integration Side

- **Mutual TLS setup in Fiorano ESB** – may require custom Java KeyStore configuration and special handling for key rotation.
 - **XML schema validation** – must ensure Fiorano ESB does not alter element ordering or namespace URIs.
 - **Asynchronous flow handling** – Fiorano ESB will need to support correlating `paymentinstruction` → `paymentinstructionresponse` messages across potentially long time spans.
 - **Message signing** – If MTN enforces XML Digital Signature, we must confirm signing algorithm and canonicalization method.
 - **Firewall rules** – Whitelisting both ways (outbound from us, inbound to MTN).
 - **Time synchronization** – Timestamps are mandatory in many requests; Fiorano must be synced via NTP to avoid transaction rejection.
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If you'd like, I can **draft a formal integration questionnaire email** that frames these questions professionally for MTN's technical team so they can respond point-by-point. This will help avoid miscommunication and speed up the integration timeline.